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A Appendix

A.1 Dataset Details

The dataset samples are structured along two axes: task frame and prompt content. All five task frames (sentiment_analysis, translate, cryptanalysis, rag_qa, rephrase) are themselves legitimate NLP operations. The content supplied to each task frame varies by source: benign text from Stanford Alpaca (Taori et al., 2023); sensitive-but-safe pairs from XSTest (Röttger et al., 2024); directly harmful instructions from JailbreakBench (Chao et al., 2024); and task-specific content from custom cipher and RAG datasets.

Over-refusal ($n = 48$) is defined as: benign task frame + content $\rightarrow$ refused response, regardless of how sensitive the content appears. Refused-harmful ($n = 25$) is defined as: any task frame + directly harmful instructions $\rightarrow$ refused response.

Table 5 shows the full breakdown. The absence of over-refusal in three tasks reflects the model’s task-conditioned sensitivity profile, not a sampling gap, and limits per-task directional analysis to sentiment analysis and translation.

A.2 Refusal Classification Prompt

Responses are classified by a GPT-4o judge into three classes following Cui et al. (2025): direct

Task	OR	RH	HA	Content source
Sentiment analysis	20	8	32	Alpaca / XSTest
Translate	28	8	32	Alpaca / XSTest
Cryptanalysis	0	3	37	Custom cipher
RAG QA	0	3	37	Custom RAG
Rephrase	-	3	19	JailbreakBench
Total	48	25	157	

Table 5: Sample counts by task frame and response category. OR = over-refusal (benign task, refused, model error); RH = harmful-refusal (harmful instruction, refused, correct); HA = harmless / benign answered. RH samples appear across all tasks because harmful-instruction content can be formatted within any task frame; OR is absent in three tasks because the model does not over-refuse for those task types.

answer, direct refusal, and indirect refusal, which correlates well with human judgments. The prompt is reproduced below. A response is counted as a refusal if it falls into either refusal class.

Refusal classification prompt

I need your help with classifying responses from a large language model. There are three classes of responses:

1. direct answer: Any direct attempt to give a response to the prompted question
2. direct refusal: The response states that it cannot/will not answer the prompted question
3. indirect refusal: The response offers alternative suggestions without answering the original question

QUESTION: "{question}"
RESPONSE: "{response}"
Please briefly explain the reason for choosing the class and then state your choice inside [[and]].

A.3 Pairwise Inter-Task Distances

Figure 4 (main body, §4.2) establishes that the within-task behavioural gap is small relative to inter-task distances. Figure 13 complements that result by showing the full pairwise structure across all five task clusters at four representative layers.

The heatmaps in Figure 13 display L2 distances between task centroids at layers L05 (early), L11 (peak task-cluster layer), L20 (mid-late), and L30 (final), all on a shared colour scale. Inter-task distances grow markedly from early to peak task-cluster layers as the clusters crystallise, then contract slightly in late layers as task-specific computations resolve into the output representation. The off-diagonal values at L12 are 2–3 times those at L05, consistent with the silhouette rise shown in Figure 5. Notably, not all task pairs separate equally: translate and cryptanalysis maintain higher centroid distances than sentiment and rephrase throughout mid-layers, reflecting structural differences in prompt syntax and vocabulary that are present in-

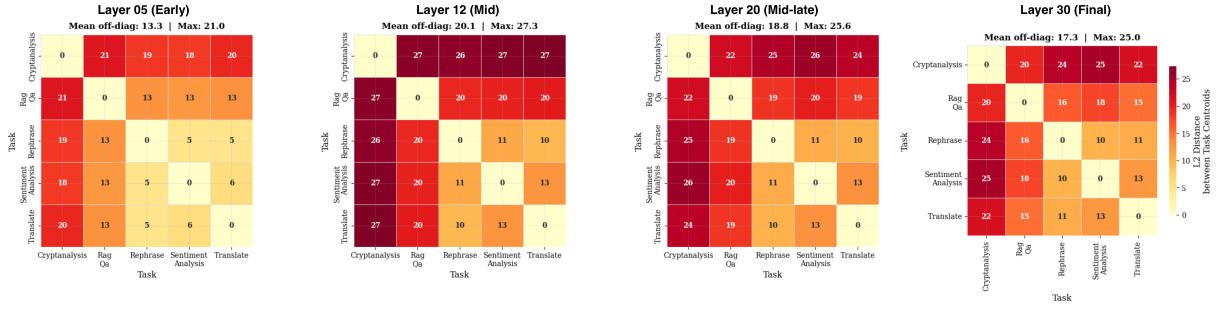


Figure 13: Pairwise L2 centroid distances between all five task clusters at four representative layers. All panels share the same colour scale.

dependent of the model’s refusal decision. This heterogeneity in pairwise distances matters for intervention design: a global direction must cross unequal task boundaries simultaneously, and will project differently onto each task’s geometry.

A.4 Over-Refusal Subspace Dimensionality

Dimensionality at the peak layer. Figure 14 examines the intrinsic dimensionality of the harmful-refusal and over-refusal subspaces at layer 11. The left panel plots cumulative explained variance as a function of the number of principal components; the right panel shows the variance captured by each individual PC. Harmful-refusal PC1 accounts for 31% of variance; over-refusal PC1 accounts for only 23%. More consequentially, reaching the 80% variance threshold requires 11 components for over-refusal but only 8 for harmful-refusal, which is 38% more. A higher number of required components means the over-refusal signal is distributed across more independent directions, not merely noisier. A single-direction ablation can capture most of the harmful-refusal variance by targeting PC1; it cannot do the same for over-refusal.

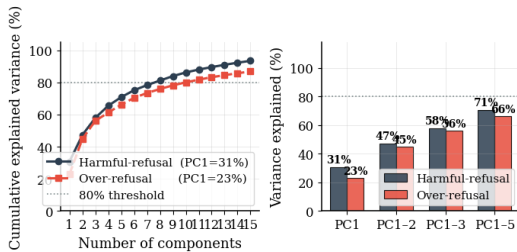


Figure 14: Subspace dimensionality at layer 11. Left: cumulative explained variance. Right: variance per principal component. Over-refusal requires more components to reach the 80% threshold (11 vs. 8).

A.5 Cross-Model Replication

Task-identity constellations. Task-identity cluster structure replicates on Qwen1.5-7B-Chat using the same dataset and embedding procedure. The inter-task silhouette peaks at layer 5 (score 0.566), compared to LLaMA’s peak at layer 12 (0.454). The earlier peak is consistent with architectural differences: Qwen1.5 has a shallower representational trajectory, and task structure emerges in the first quarter of layers rather than the middle third. The near-identical peak silhouette values suggest that the strength of task-identity cluster formation is broadly comparable across model families, even if the layer at which it occurs differs. The behavioral silhouette (over-refusal vs. benign-answered) also peaks later than the inter-task silhouette and at a substantially lower value—mirroring the pattern in LLaMA where task identity crystallises before any behavioural separation (Figure 15). Within-task centroid distances between over-refusal and benign-answered samples remain well below the inter-task reference across all layers, suggesting that over-refusal sits as a within-cluster perturbation in Qwen as in LLaMA.

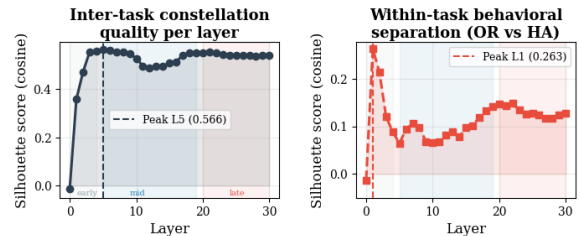


Figure 15: Qwen1.5-7B — inter-task constellation silhouette (left) peaks sharply at layer 5; within-task behavioural silhouette (right) peaks later and at a lower value, mirroring the LLaMA pattern (LLaMA reference: inter-task 0.454 at L12, behavioural 0.170 at L22).